

Hofstadter spectrum of TBG

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June 15, 2026

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1 BM Hamiltonian

The lattice vectors of TBG in real and reciprocal spaces are respectively (in basis $\mathbf{e}_x, \mathbf{e}_y$)

$$\mathbf{a}_1 = L_\theta \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right), \quad \mathbf{a}_2 = L_\theta(0, 1); \quad \mathbf{b}_1 = \frac{4\pi}{\sqrt{3}L_\theta}(1, 0), \quad \mathbf{b}_2 = \frac{4\pi}{\sqrt{3}L_\theta} \left(-\frac{1}{2}, \frac{\sqrt{3}}{2} \right). \quad (1)$$

Under external field $\mathbf{B} = B_z \mathbf{e}_z = \chi B \mathbf{e}_z$ ($B > 0$, $\chi = \pm 1$), the BM Hamiltonian reads ($\boldsymbol{\sigma}_\eta = (\eta\sigma_x, \sigma_y)$)

$$H^\eta = \begin{pmatrix} v_F(\boldsymbol{\pi} - \hbar \mathbf{K}_1^\eta) \cdot \boldsymbol{\sigma}_\eta & U_\eta(\mathbf{r}) \\ U_\eta^\dagger(\mathbf{r}) & v_F(\boldsymbol{\pi} - \hbar \mathbf{K}_2^\eta) \cdot \boldsymbol{\sigma}_\eta \end{pmatrix}, \quad (2)$$

where $\boldsymbol{\pi} = \mathbf{p} + e\mathbf{A}$ in the Landau gauge $\mathbf{A} = \chi B x \mathbf{e}_y$. The moiré potential is

$$U_\eta(\mathbf{r}) = T_1^\eta + T_2^\eta e^{-i\eta \mathbf{b}_2 \cdot \mathbf{r}} + T_3^\eta e^{-i\eta(\mathbf{b}_1 + \mathbf{b}_2) \cdot \mathbf{r}}, \quad (3)$$

with

$$T_1^\eta = \begin{pmatrix} u_0 & u_1 \\ u_1 & u_0 \end{pmatrix}, \quad T_2^\eta = \begin{pmatrix} u_0 & u_1 \omega^\eta \\ u_1 \omega^{-\eta} & u_0 \end{pmatrix}, \quad T_3^\eta = \begin{pmatrix} u_0 & u_1 \omega^{-\eta} \\ u_1 \omega^\eta & u_0 \end{pmatrix}. \quad (4)$$

The Dirac point in layer l is

$$\mathbf{K}_l^\eta = \eta(\boldsymbol{\Gamma} + \bar{\mathbf{K}}_l), \quad \bar{\mathbf{K}}_1 = \frac{4\pi}{3L_\theta} \left(\frac{\sqrt{3}}{2}, -\frac{1}{2} \right), \quad \bar{\mathbf{K}}_2 = \frac{4\pi}{3L_\theta} \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right), \quad (5)$$

where the momentum shift $\boldsymbol{\Gamma}$ can be gauged out.

We focus on the case with signed flux $\Phi/\Phi_0 = \chi p/q$, where

$$\Phi = \chi B \frac{\sqrt{3}L_\theta^2}{2}, \quad \Phi_0 = h/e, \quad (6)$$

or equivalently $|\Phi|/\Phi_0 = p/q$. Thus

$$\frac{\sqrt{3}L_\theta^2/2}{2\pi l_B^2} = \frac{p}{q} \quad \Rightarrow \quad l_B = L_\theta \sqrt{\frac{\sqrt{3}}{4\pi} \frac{q}{p}}, \quad (7)$$

where $l_B = \sqrt{\hbar/(eB)}$ is the magnetic length.

2 LL basis for graphene

We use the LL basis of moiré-less monolayer graphene as the starting point. The monolayer graphene has the Hamiltonian (omit the momentum offset)

$$h_l^\eta(\boldsymbol{\pi}) = v_F \boldsymbol{\pi} \cdot \boldsymbol{\sigma}_\eta = i \frac{\hbar v_F}{\sqrt{2} l_B} \begin{pmatrix} 0 & (\eta - \chi) a^\dagger - (\eta + \chi) a \\ (\eta + \chi) a^\dagger + (\chi - \eta) a & 0 \end{pmatrix}. \quad (8)$$

Both valleys have LL energies $\epsilon_n = \text{sgn}(n) \sqrt{2|n|} \hbar v_F / l_B$. If $\eta = -\chi$, the LL wave functions are

$$\tilde{\phi}_{n=0, \mathbf{k}}^\eta = \begin{pmatrix} \psi_{0\mathbf{k}}^\chi \\ 0 \end{pmatrix}, \quad \tilde{\phi}_{n \neq 0, \mathbf{k}}^\eta = \frac{1}{\sqrt{2}} \begin{pmatrix} \psi_{|n|\mathbf{k}}^\chi \\ i\chi \text{sgn}(n) \psi_{|n|-1, \mathbf{k}}^\chi \end{pmatrix}, \quad (9)$$

while if $\eta = \chi$, the LL wave functions are

$$\tilde{\phi}_{n=0, \mathbf{k}}^\eta = \begin{pmatrix} 0 \\ \psi_{0\mathbf{k}}^\chi \end{pmatrix}, \quad \tilde{\phi}_{n \neq 0, \mathbf{k}}^\eta = \frac{1}{\sqrt{2}} \begin{pmatrix} -i\chi \text{sgn}(n) \psi_{|n|-1, \mathbf{k}}^\chi \\ \psi_{|n|\mathbf{k}}^\chi \end{pmatrix}. \quad (10)$$

Here $\psi_{n\mathbf{k}}^\chi$ are LL wave functions of quadratic electrons on torus,

$$\begin{aligned} \psi_{n\mathbf{k}}^\chi &= \frac{e^{i\chi\pi k_2^2 \frac{m_1 y}{m_2}}}{\sqrt{N_1 N_t}} \sum_{s=0}^{N_1-1} \sum_{l \in \mathbb{Z}} e^{i2\pi(s+lN_1)k_1 t^{s+lN_1}} (\mathbf{m}_1) \varphi_{n, \frac{2\pi}{m_2} k_2} \\ &= \frac{1}{\sqrt{N_1 N_t}} \sum_{s \in \mathbb{Z}} e^{is2\pi k_1} e^{i\chi\pi \frac{m_1 y}{m_2} (k_2 - \chi s)^2} \varphi_{n, \frac{2\pi}{m_2} (k_2 - \chi s)}, \end{aligned} \quad (11)$$

and φ_{n, k_y} are LLs on cylinder. From Eqs. (9), (10), it is important to keep the LL cutoff on the zero-mode sublattice larger by one. Namely, the A -sublattice cutoff should be larger when $\eta = -\chi$, while the B -sublattice cutoff should be larger when $\eta = \chi$.

3 TBG matrix elements

3.1 Basis set

For rational flux $|\Phi|/\Phi_0 = p/q$, we use the $q\mathbf{a}_1$ convention of the continuum Hofstadter note. The flux-one magnetic lattice in real and reciprocal space is

$$\mathbf{m}_1 = \frac{q}{p} \mathbf{a}_1, \quad \mathbf{m}_2 = \mathbf{a}_2, \quad \mathbf{g}_1 = \frac{p}{q} \mathbf{b}_1, \quad \mathbf{g}_2 = \mathbf{b}_2. \quad (12)$$

We also define the commensurate supercell vectors and reciprocal vectors

$$\mathbf{L}_1 = q\mathbf{a}_1 = p\mathbf{m}_1, \quad \mathbf{L}_2 = \mathbf{a}_2 = \mathbf{m}_2, \quad (13)$$

$$\mathbf{P}_1 = \frac{\mathbf{b}_1}{q} = \frac{\mathbf{g}_1}{p}, \quad \mathbf{P}_2 = \mathbf{b}_2 = \mathbf{g}_2. \quad (14)$$

The momentum $\tilde{\mathbf{k}} \in \text{MBZ}$ can be parameterized as

$$\tilde{\mathbf{k}} = \mathbf{k} + r\mathbf{P}_1 = (k_1 + r)\mathbf{P}_1 + k_2\mathbf{P}_2 = \frac{k_1 + r}{p} \mathbf{g}_1 + k_2 \mathbf{g}_2 = \frac{k_1 + r}{q} \mathbf{b}_1 + k_2 \mathbf{b}_2, \quad (15)$$

with $\mathbf{k} = k_1\mathbf{P}_1 + k_2\mathbf{P}_2 \in \text{SBZ}$, $k_1, k_2 \in [0, 1)$, and $r = 0, 1, \dots, p-1$. Following note ‘Hofstadter’ and gauging out the offset $\boldsymbol{\Gamma}$, we define the following basis

$$|\eta l \alpha n r \mathbf{k}\rangle = e^{i\eta \boldsymbol{\Gamma} \cdot \mathbf{r}} \chi_{l\alpha} |\psi_{n, \frac{k_1+r}{p} \mathbf{g}_1 + k_2 \mathbf{g}_2}^\chi\rangle = e^{i\eta \boldsymbol{\Gamma} \cdot \mathbf{r}} \chi_{l\alpha} |\psi_{n, \frac{k_1+r}{q} \mathbf{b}_1 + k_2 \mathbf{b}_2}^\chi\rangle, \quad (16)$$

where $\chi_{l\alpha}$ is a 4-vector whose $(l\alpha)$ element is 1 and 0 for others. The plane wave matrix elements are derived in note ‘Hofstadter’, so we directly list the Hamiltonian matrix elements below, which are diagonal in $\mathbf{k}$.

3.2 Matrix elements in + valley

First focus on the valley $\eta = +$. The kinetic term is diagonal in l, r , with elements

$$\begin{aligned} & \langle +l\alpha'n'r\mathbf{k} | v_F(\boldsymbol{\pi} - \hbar\mathbf{K}_l^+) \cdot \boldsymbol{\sigma}_+ | +l\alpha n r \mathbf{k} \rangle \\ = & i \frac{\hbar v_F}{\sqrt{2}l_B} \begin{pmatrix} 0 & (1-\chi)\sqrt{n'}\delta_{n',n+1} - (1+\chi)\sqrt{n}\delta_{n'+1,n} \\ (1+\chi)\sqrt{n'}\delta_{n',n+1} + (\chi-1)\sqrt{n}\delta_{n'+1,n} & 0 \end{pmatrix}_{\alpha'\alpha} - \hbar v_F [\bar{\mathbf{K}}_l \cdot \boldsymbol{\sigma}_+]_{\alpha'\alpha} \delta_{n'n}. \end{aligned} \quad (17)$$

For the moiré potential,

$$\langle +1\alpha'n'r\mathbf{k} | U_+ | +2\alpha n r \mathbf{k} \rangle = [T_1^+]_{\alpha'\alpha} \delta_{n'n} \delta_{r'r} + [T_2^+]_{\alpha'\alpha} [e^{-i\mathbf{b}_2 \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi + [T_3^+]_{\alpha'\alpha} [e^{-i(\mathbf{b}_1+\mathbf{b}_2) \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi, \quad (18)$$

where

$$[e^{-i\mathbf{b}_2 \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi = F_{n'n} (Q_\chi(-\mathbf{b}_2)) e^{-i\chi \frac{2\pi}{p}(k_1+r)} \delta_{r'r}, \quad (19)$$

$$[e^{-i(\mathbf{b}_1+\mathbf{b}_2) \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi = F_{n'n} (Q_\chi(-\mathbf{b}_1 - \mathbf{b}_2)) e^{-i\chi \frac{2\pi}{p}(k_1+r) + i\chi \frac{2\pi q}{p}(k_2+\frac{1}{2})} \delta_{r',r-q}^{\text{mod } p}. \quad (20)$$

Here

$$Q_\chi(\mathbf{q}) = \frac{(q_x + i\chi q_y)l_B}{\sqrt{2}}, \quad (21)$$

and the form factor reads

$$F_{n'n}(Q) = \begin{cases} e^{-\frac{|Q|^2}{2}} \sqrt{\frac{n!}{n'}} (iQ)^{n-n'} L_{n'-n}^{n-n'}(|Q|^2) & (n' \leq n) \\ e^{-\frac{|Q|^2}{2}} \sqrt{\frac{n!}{n'}} (iQ^*)^{n'-n} L_n^{n'-n}(|Q|^2) & (n' \geq n) \end{cases}. \quad (22)$$

3.3 Matrix elements in - valley

In the valley $\eta = -$, the kinetic term is

$$\begin{aligned} & \langle -l\alpha'n'r\mathbf{k} | v_F(\boldsymbol{\pi} - \hbar\mathbf{K}_l^-) \cdot \boldsymbol{\sigma}_- | -l\alpha n r \mathbf{k} \rangle \\ = & i \frac{\hbar v_F}{\sqrt{2}l_B} \begin{pmatrix} 0 & (-1-\chi)\sqrt{n'}\delta_{n',n+1} + (1-\chi)\sqrt{n}\delta_{n'+1,n} \\ (-1+\chi)\sqrt{n'}\delta_{n',n+1} + (\chi+1)\sqrt{n}\delta_{n'+1,n} & 0 \end{pmatrix}_{\alpha'\alpha} + \hbar v_F [\bar{\mathbf{K}}_l \cdot \boldsymbol{\sigma}_-]_{\alpha'\alpha} \delta_{n'n}. \end{aligned} \quad (23)$$

The moiré potential is

$$\langle -1\alpha'n'r\mathbf{k} | U_- | -2\alpha n r \mathbf{k} \rangle = [T_1^-]_{\alpha'\alpha} \delta_{n'n} \delta_{r'r} + [T_2^-]_{\alpha'\alpha} [e^{i\mathbf{b}_2 \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi + [T_3^-]_{\alpha'\alpha} [e^{i(\mathbf{b}_1+\mathbf{b}_2) \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi, \quad (24)$$

where

$$[e^{i\mathbf{b}_2 \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi = F_{n'n} (Q_\chi(\mathbf{b}_2)) e^{i\chi \frac{2\pi}{p}(k_1+r)} \delta_{r'r}, \quad (25)$$

$$[e^{i(\mathbf{b}_1+\mathbf{b}_2) \cdot \mathbf{r}}]_{n'r'\mathbf{k},nr\mathbf{k}}^\chi = F_{n'n} (Q_\chi(\mathbf{b}_1 + \mathbf{b}_2)) e^{i\chi \frac{2\pi}{p}(k_1+r) - i\chi \frac{2\pi q}{p}(k_2-\frac{1}{2})} \delta_{r',r+q}^{\text{mod } p}. \quad (26)$$

4 Chern number

For the Hofstadter state $|\Phi_{\nu\mathbf{k}}^\eta\rangle = \sum_{l\alpha n r} |\eta l\alpha n r \mathbf{k}\rangle P_{l\alpha n r, \nu}^\eta(\mathbf{k})$ satisfying $H^\eta |\Phi_{\nu\mathbf{k}}^\eta\rangle = E_{\nu\mathbf{k}}^\eta |\Phi_{\nu\mathbf{k}}^\eta\rangle$, we define the Bloch wave function $|u_{\nu\mathbf{k}}^\eta\rangle = e^{-i\mathbf{k} \cdot \mathbf{r}} |\Phi_{\nu\mathbf{k}}^\eta\rangle$, where $\mathbf{k} = k_1 \mathbf{P}_1 + k_2 \mathbf{P}_2 = \frac{k_1}{p} \mathbf{g}_1 + \frac{k_2}{p} \mathbf{g}_2 \in \text{SBZ}$. The (valley) Chern number of band ν can be calculated as follows. We discretize $\mathbf{k}_{ij} = (\frac{i}{N_1}, \frac{j}{N_2})$, $i = 0, 1, \dots, N_1 - 1$, $j = 0, 1, \dots, N_2 - 1$, and define

$$U_1^{\eta\nu}(\mathbf{k}_{ij}) = \frac{\langle u_{\nu, \mathbf{k}_{ij}}^\eta | u_{\nu, \mathbf{k}_{i+1, j}}^\eta \rangle}{|\langle u_{\nu, \mathbf{k}_{ij}}^\eta | u_{\nu, \mathbf{k}_{i+1, j}}^\eta \rangle|}, \quad U_2^{\eta\nu}(\mathbf{k}_{ij}) = \frac{\langle u_{\nu, \mathbf{k}_{ij}}^\eta | u_{\nu, \mathbf{k}_{i, j+1}}^\eta \rangle}{|\langle u_{\nu, \mathbf{k}_{ij}}^\eta | u_{\nu, \mathbf{k}_{i, j+1}}^\eta \rangle|}, \quad \mathcal{F}^{\eta\nu}(\mathbf{k}_{ij}) = \ln \left[\frac{U_1^{\eta\nu}(\mathbf{k}_{ij}) U_2^{\eta\nu}(\mathbf{k}_{i+1, j})}{U_1^{\eta\nu}(\mathbf{k}_{i, j+1}) U_2^{\eta\nu}(\mathbf{k}_{ij})} \right], \quad (27)$$

where the branch of $\mathcal{F}$ is chosen such that $\text{Im } \mathcal{F} \in [-\pi, \pi)$. The valley Chern number reads

$$\sigma_\nu^\eta = \frac{1}{2\pi i} \sum_{ij} \mathcal{F}^{\eta\nu}(\mathbf{k}_{ij}). \quad (28)$$

The inner product of $|u_{\nu\mathbf{k}}^\eta\rangle$ reads

$$\langle u_{\nu'\mathbf{k}'}^\eta | u_{\nu\mathbf{k}}^\eta \rangle = \sum_{l\alpha n'r'} \sum_{nr} P_{l\alpha n'r',\nu'}^{\eta*}(\mathbf{k}') [e^{i(\mathbf{k}'-\mathbf{k})\cdot\mathbf{r}}]_{n'r'\mathbf{k}',nr\mathbf{k}} P_{l\alpha nr,\nu}^\eta(\mathbf{k}) = [P^{\eta\dagger}(\mathbf{k}') X^\chi(\mathbf{k}', \mathbf{k}) P^\eta(\mathbf{k})]_{\nu'\nu}, \quad (29)$$

where

$$[X^\chi(\mathbf{k}', \mathbf{k})]_{l'\alpha'n'r',l\alpha nr} = \delta_{l'l} \delta_{\alpha'\alpha} \delta_{r'r} e^{-i\chi \frac{\pi}{p} (k'_1 - k_1)(k'_2 + k_2)} F_{n'n}(Q_\chi(\mathbf{k}' - \mathbf{k})), \quad (30)$$

with $\mathbf{k}' - \mathbf{k} = (k'_1 - k_1)\mathbf{P}_1 + (k'_2 - k_2)\mathbf{P}_2 = \frac{k'_1 - k_1}{p}\mathbf{g}_1 + (k'_2 - k_2)\mathbf{g}_2$. For collective bands, we only need to replace U by the multi-band determinant. When a link crosses the SBZ boundary, $\mathbf{k}'$ in $X^\chi(\mathbf{k}', \mathbf{k})$ should be treated as the lifted momentum before reduction modulo the SBZ. The eigenvector at the reduced momentum must then be related to the lifted state by the corresponding boundary transition function.